

An Internship Report
Of
Python Fullstack Developer

Submitted

To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B-Tech (CSE AI/ML)



By

Roll Number of Student:- 2410030715

Name of Student: RAJYAVARDHAN

Batch: 2026-27

1. CANDIDATE'S DECLARATION

I, RAJYAVARDHAN do hereby declare that the internship report titled “**Python Fullstack Developer**” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature

Student Name:- **RAJYAVARDHAN**

Roll No.:- **2410030715**

2. ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to thank the **AICTE – EduSkills Virtual Internship Program team and EduSkills Academy** for providing me the opportunity and the structured platform to learn and gain hands-on exposure in the domain of **“Python Fullstack Developer”**. I am also thankful to my institute, IILM University, Greater Noida, and the faculty of the School of Computer Science and Engineering for their continuous support and encouragement.

I express my gratitude to my parents for their blessings.

Signature

Student Name:- **RAJYAVARDHAN**

Roll No.:- **2410030715**

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3. Internship Completion Certificate

**CERTIFICATE
OF INTERNSHIP**





PRESENTED TO

Rajyavardhan .

has successfully achieved the Internship Credential by completing a 8-week program in the domain of

Python Full Stack

During the internship, the learner demonstrated proficiency in:

- HTML: Foundations of Web Development
- CSS: Styling and Designing Web Pages & Bootstrap: Responsive Web Design Framework
- JavaScript Fundamentals and DOM Manipulation
- jQuery: Simplifying JavaScript Development
- Python: Programming Fundamentals and Applications
- Building Web Applications Using Django
- SQL: Database Querying and Management
- Git & Version Control: Managing Code Effectively

This certification recognizes the successful completion of the internship requirements.



Issued on: Jul 29, 2026
Certificate ID: 2026-7B1659F7D5



Mitu Swain
General Manager
(Learning & Development)
EduSkills



शिक्षा मंत्रालय
MINISTRY OF
EDUCATION



अखिल भारतीय तकनीकी शिक्षा परिषद्
All India Council for Technical Education



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Certificate of Virtual Internship

This is to certify that

Rajyavardhan .

IILM University, Greater Noida

has successfully completed the 8-weeks

Python Fullstack Developer Virtual Internship

During June - August 2026

Supported by



EduSkills
ACADEMY

Dr. Buddha Chandrasekhar
Chief Coordinating Officer (CCO)
AICTE, Ministry of Education

Shubhajit Jagadev
Chief Executive Officer (CEO)
EduSkills



Certificate ID : 44478c1afe30f54e3097

Student ID: STU6989ebf52c9a81770646517



GRADE- O (Outstanding):90-100 | E (Excellent):80-89 | A (Very Good):70-79 | B (Good): 60-69 | C (Fair): 50-59 | D (Average): 40-49 | P (Pass): 30-39 | F (Fail): Below 30

Internship Offer Letter



Ref No: C17Y2026M081452593

Internship Offer Letter

Student Name : Rajyavardhan .
Stream : B.Tech
Branch : B.Tech(Computer Science and Engineering)
Institute Name : IILM University, Greater Noida
Internship Domain : Python Fullstack Developer
Internship Duration : June 2026 - August 2026
AICTE Student ID : STU6989ebf52c9a81770646517

We are pleased to inform you that you have been selected for the 8-Week **AICTE – EduSkills Virtual Internship Program**.

During this internship, you will learn and demonstrate industry-relevant skills that will enhance your employability and strengthen your confidence in the subject area.

The program is designed to provide structured learning combined with practical exposure.

Internship Structure & Evaluation:

- The internship will follow a **week-wise structured learning plan**.
- You will be required to complete **weekly assessments** to track your progress and understanding.
- Certain modules will require submission of **project documentation and assignments**.
- In the final week, you must appear for a **Final Assessment Test**.
- The final grade will be based on your overall performance, including weekly assessments, project submissions, and the final test.

Upon successful completion of all required steps and assessments, you will be awarded a **Virtual Internship Certificate**.

Please note that this offer letter will be considered valid only after the successful completion of the internship program and fulfillment of all assessment requirements, including obtaining the final completion certificate. Without the final completion certificate, this offer letter shall not be treated as a valid document for any official purpose.

The week wise Structure of the Internship is as Follows:

Week	Module	Description
Week 1	HTML	Learn HyperText Markup Language (HTML), including page structure, elements, forms, tables, semantic tags, and website layout fundamentals.
Week 2	CSS	Master Cascading Style Sheets (CSS) including selectors, box model, positioning, flexbox, grid, responsiveness, and styling techniques.
Week 3	Bootstrap	Build responsive and mobile-first websites using Bootstrap components, grid system, utilities, and pre-built UI elements.
Week 4	JavaScript	Learn JavaScript fundamentals including variables, functions, loops, conditionals, DOM manipulation, and event handling.
Week 5	jQuery	Simplify JavaScript development using jQuery for DOM traversal, animations, event handling, and AJAX interactions.
Week 6	Python Programming	Learn Python basics including syntax, data types, loops, functions, OOP concepts, and backend development fundamentals.
Week 7	Django Framework	Build dynamic web applications using Django with models, views, templates, routing, and authentication systems.
Week 8	MySQL & Database Management	Learn SQL fundamentals, database design, CRUD operations, and integrate MySQL with Django applications.
Internship Grade Point Assessment		

Note: There is no stipend associated with this internship.

We congratulate you on your selection and wish you a successful learning journey.



4. PROJECT DESCRIPTION

4.1 Introduction

This report presents a detailed account of the 8-week Virtual Internship undertaken under the AICTE – EduSkills Virtual Internship Program, in the domain of **“Python Full Stack Developer”**. The internship was carried out from June 2026 to August 2026 as part of the requirement of the Bachelor of Technology (Computer Science and Engineering) programme at IILM University, Greater Noida. The primary aim of this internship was to bridge the gap between theoretical understanding of software development concepts and their practical implementation in real-world web applications. Over the course of the programme, the learner was exposed to industry-relevant tools and technologies used in developing dynamic, responsive, and database-driven web applications.

The internship followed a structured, week-wise curriculum that combined conceptual learning with hands-on assignments, weekly assessments, project documentation, and a final capstone project. Throughout the programme, emphasis was placed on frontend development using HTML, CSS, and JavaScript, backend development using Python, and database integration. The learner gained practical exposure to designing user interfaces, implementing application logic, managing data, and understanding the software development lifecycle. On successful completion of all the requirements, including the final assessment test, a Virtual Internship Certificate was awarded by EduSkills Academy.

4.2 Organization Profile

EduSkills Academy is an ISO 9001:2015 and ISO 20000-1:2018 certified organization working under the theme “Nation Building Through Skills”. EduSkills partners with the All India Council for Technical Education (AICTE) to run the AICTE – EduSkills Virtual Internship Program under the patronage of the AICTE National Internship Portal. The organization collaborates with leading technology companies and academic institutions to design industry-aligned virtual internship curricula that help engineering students strengthen their employability and gain practical exposure to emerging technology domains such as Artificial Intelligence, Cloud Computing, Cybersecurity, Software Development, Full Stack Development, and Data Science. Through its industry-oriented training programs, EduSkills enables students to develop practical skills and gain real-world experience required for successful professional careers.

I am also thankful to my university, IILM University, Greater Noida, for encouraging students to participate in such industry-oriented learning programs. The knowledge and experience gained during this internship will be highly beneficial for my academic growth and future professional career.

4.3 Problem Statement

Although many engineering students learn programming languages, web technologies, and database concepts during their academic coursework, only a limited number gain practical exposure to developing complete web applications that meet industry standards. Understanding individual technologies such as HTML, CSS, JavaScript, Python, and databases is only one part of modern software development; integrating these technologies to build responsive, secure, scalable, and user-friendly applications is an equally important skill. In today's digital era, organizations require developers who can work across both frontend and backend technologies and contribute to the complete software development lifecycle. The problem addressed by this internship was to understand and practically implement the complete full-stack development process – from designing interactive user interfaces and developing backend functionality to database integration, application testing, deployment, and maintenance of real-world web applications. The internship aimed to provide practical exposure to industry-oriented development practices and bridge the gap between academic learning and professional software development.

4.4 Project Objectives

- To understand the complete full-stack web development lifecycle and industry-standard development methodologies used in software projects.
- To learn frontend development using HTML, CSS, and JavaScript for creating responsive, interactive, and user-friendly web interfaces.
- To gain hands-on experience in backend development using Python and web development frameworks for building dynamic applications.
- To understand database design, data storage, database connectivity, and CRUD operations using relational database management systems.
- To develop dynamic and database-driven web applications by integrating frontend, backend, and database components effectively.
- To learn application testing, debugging, error handling, and performance optimization techniques for reliable software development.
- To understand version control systems and collaborative development practices commonly used in modern software engineering teams.
- To implement authentication, authorization, security, and data validation mechanisms to ensure application reliability and data protection.
- To gain practical exposure to application deployment, maintenance, and real-world software development workflows.

4.5 Scope of the Project

The scope of this internship spans the complete full-stack web development lifecycle – starting from designing responsive user interfaces, implementing frontend functionality, developing backend services using Python, integrating databases, and testing application features. It also covers the development of dynamic and database-driven web applications that provide an interactive user experience while ensuring security, scalability, and maintainability. The internship focused on understanding how frontend, backend, and database components work together to create real-world web solutions. The scope further included application deployment, debugging, version control practices, and performance optimization techniques used in modern software development. The internship was limited to a virtual learning and development environment provided by EduSkills Academy and did not involve deployment on any live organizational production system.

4.6 Technologies and Tools Used

- Programming Language: Python
- Frontend Development: HTML, CSS, JavaScript
- Backend Development: Python, Flask
- Database Management: MySQL
- Development Environment: Visual Studio Code
- Version Control: Git and GitHub
- Web Technologies: RESTful APIs, Client-Server Architecture
- Testing & Debugging: Browser Developer Tools, Python Debugging Utilities
- Deployment & Hosting: Basic Web Application Deployment Techniques

4.7 System Architecture

The general architecture followed during the internship for developing a full-stack web application can be summarized in the following layers:

- Presentation Layer: The user interface is developed using HTML, CSS, and JavaScript to provide an interactive and responsive experience for end users.
- Frontend Layer: Client-side scripts handle user interactions, form validations, page rendering, and communication with backend services.

- **Application Layer:** The backend application is developed using Python and Flask, where business logic, request processing, and application workflows are implemented.
- **API Layer:** RESTful APIs are used to facilitate communication between the frontend interface and backend services, ensuring efficient data exchange.
- **Database Layer:** MySQL is used for storing, retrieving, updating, and managing application data securely and efficiently.
- **Integration Layer:** The frontend, backend, and database components are integrated to ensure seamless functionality and smooth user interaction across the application.
- **Development & Version Control Layer:** Git and GitHub are used for source code management, version tracking, and collaborative development practices.
- **Deployment Layer:** The completed application is tested, optimized, and deployed in a development environment to ensure accessibility, reliability, and performance.

4.8 Methodology

The internship followed a structured, week-wise methodology combining conceptual learning, hands-on labs, coding exercises, weekly assessments, and project submissions, as summarized in the table below.

Week	Module	Description
1.	Introduction to Full Stack Development & Web Fundamentals	Introduction to full-stack development, software development lifecycle, client-server architecture, web technologies, environment setup, and fundamentals of HTML and CSS.
2.	Frontend Development with HTML, CSS & JavaScript	Building responsive web pages, styling with CSS, JavaScript fundamentals, DOM manipulation, form handling, and interactive user interface development.
3.	Advanced Frontend Development	Responsive design techniques, Bootstrap framework, advanced JavaScript concepts, event handling, validation and improving user experience

4.	Python Programming for Web Development	Python fundamentals, functions, modules, object-oriented programming concepts, exception handling, file operations, and application development basics.
5.	Backend Development with Python & Flask	Introduction to Flask, routing, templates, request handling, session management, backend logic implementation, and REST API development.
6.	Database Integration & CRUD Operations	Database concepts, MySQL connectivity, data modeling, CRUD operations, SQL queries, data validation, and backend-database integration.
7.	Full Stack Application Development & Testing	Integrating frontend, backend, and database components, debugging, testing, performance optimization, version control using Git and GitHub, and deployment concepts.
8.	Capstone Project & Application Deployment	Development of a complete end-to-end full-stack web application, implementation of core features, testing, deployment, documentation, and project presentation.

4.9 Expected Outcomes

- To gain the ability to independently design, develop, and deploy dynamic full-stack web applications using Python and modern web technologies.
- To develop practical understanding of frontend development using HTML, CSS, JavaScript, and responsive design principles for creating user-friendly interfaces.
- To acquire hands-on experience in backend development using Python and web frameworks for implementing business logic and application functionality.
- To gain proficiency in database design, SQL queries, CRUD operations, and integration of databases with web applications.
- To understand the complete software development lifecycle, including requirement analysis, development, testing, debugging, deployment, and maintenance.
- To develop the ability to implement authentication, authorization, form validation, and security mechanisms in web applications.
- To gain practical exposure to version control systems, collaborative development practices, and project documentation using industry-standard tools.
- To enhance problem-solving, debugging, and analytical skills required for professional software development.
- To gain experience in developing RESTful APIs and integrating them with frontend applications for seamless data communication.
- To understand application deployment concepts, server configuration, and best practices for hosting web applications.
- To improve teamwork, project management, and technical communication skills through project-based learning activities.
- To complete an end-to-end capstone project demonstrating the integration of frontend, backend, and database technologies in a real-world full-stack web application.

4.10 Certificates of Completion and Communication Proof

The Internship Offer Letter and the Internship Completion Certificate issued by EduSkills Academy, confirming successful completion of the 10-weeks AICTE – EduSkills Virtual Internship Program in **Python FullStack Developer**, are attached in Chapter 3 of this report.

8/25/26, 7:26 PM

IILM Mail - AICTE-EduSkills Internship Certificate download link



Rajyavardhan <rajya.vardhan.cs28@iilm.edu>

AICTE-EduSkills Internship Certificate download link

1 message

EduSkills Foundation <noreply@eduskillsfoundation.org>
To: "Rajyavardhan ." <rajya.vardhan.cs28@iilm.edu>

Wed, Jul 29, 2026 at 6:57 AM

Dear **Rajyavardhan** .,

Congratulations!

You have successfully completed **10 weeks Python Fullstack Developer**. Your certificate is ready — please log in to your profile and download it.

Steps to download your certificate:

1. Login at: <https://eduskillsfoundation.org/login>
2. Go to the **Certificate Section** from your sidebar
3. Click **Download Certificate**

Please post your certificate on your LinkedIn profile to get better job opportunities and visibility from corporates.

Why You Should Share Your Achievements on LinkedIn

- 1. World's Largest Professional Platform**
LinkedIn has **900+ million professionals** and **60+ million companies** worldwide.
- 2. Recruiters Actively Use LinkedIn**
Around **8 out of 10 recruiters** use LinkedIn to find and verify candidates.
- 3. Profiles Are Checked Before Hiring**
More than **90% of hiring managers** review LinkedIn profiles before shortlisting candidates.
- 4. Better Visibility for Freshers**
Students who post **certificates, internships, and projects** appear in recruiter searches more often.
- 5. Skill-Based Hiring is Growing**
Employers now focus more on **skills, certifications, and practical experience**.
- 6. Higher Chances of Job Opportunities**
Active LinkedIn profiles with achievements get **more internship and job opportunities**.

<https://mail.google.com/mail/u/0/?ik=16eeb6f452&view=pt&search=all&permthid=thread-f:1872010597249532090&simpl=msg-f:1872010597249532090> 1/2

5. BIBLIOGRAPHY/REFERENCES

1. EduSkills Academy, “AICTE – EduSkills Virtual Internship Program – Python FullStack Developer,” Internship Offer Letter, Ref No. C17Y2026M081452593, 2026.
2. EduSkills Academy, “Certificate of Internship – Python FullStack Developer,” Certificate ID: 44478c1afe30f54e3097, Issued on July 29, 2026.